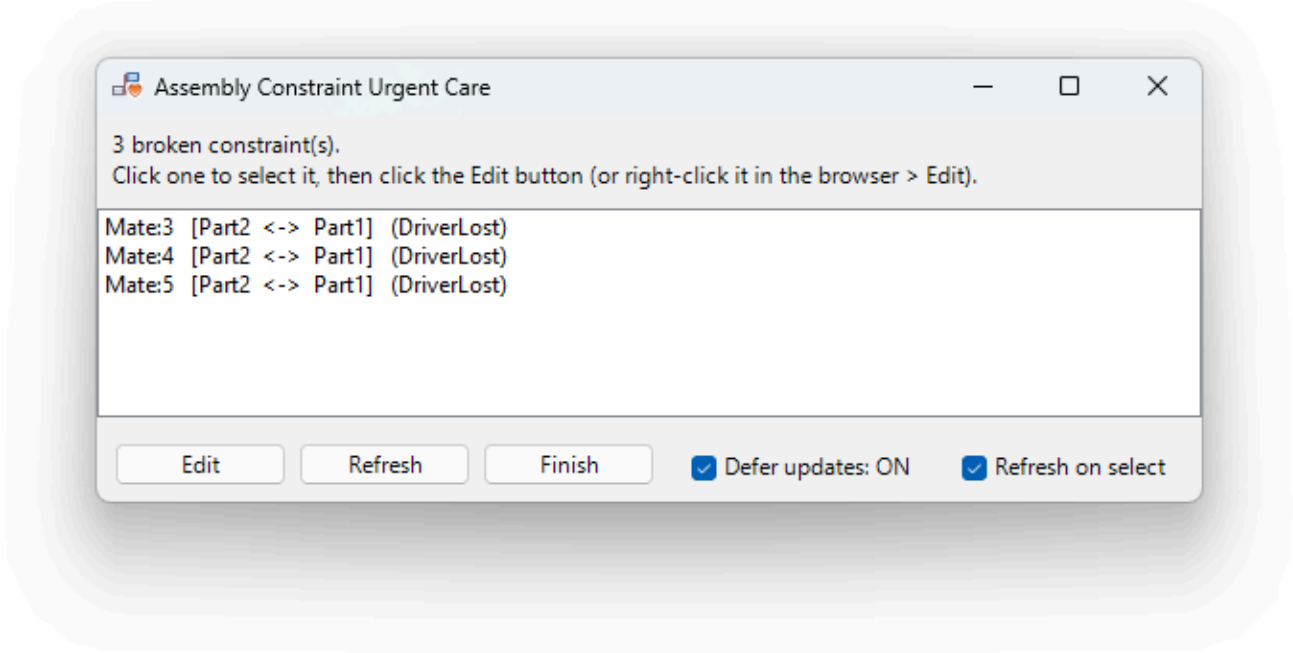
Assembly Constraint Urgent Care

A small add-in for Autodesk Inventor that lists the broken ("sick") constraints in an assembly in one window, lets you pick one to select and edit, and turns Defer Update on and off for you while you fix them.

Works with Inventor 2025, 2026 and 2027 (Windows, 64-bit). Developed on 2025.4 and tested on 2025 and 2027. The add-in carries its own copy of the Inventor interop, so the same DLL runs on all three. Later versions have not been tested.

Where to find it: the add-in installs a **Constraint Urgent Care** button on the **Assemble** tab, **Relationships** panel, in the Assembly environment.

Download and source code: https://github.com/CurtisWagu/Assembly_Constraint_Urgent_Care



> **Provided as is. Not supported.** See Disclaimer.

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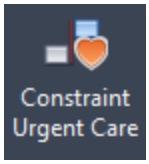
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What it does

When an assembly has several broken constraints, fixing one can make Inventor show its constraint warning again, even though the one you just fixed is fine and the problem is in the others. This add-in gives you one place to work through them.

- Adds a **Constraint Urgent Care** button to the **Relationships** panel on the **Assemble** tab, in the Assembly environment.



- Lists every broken constraint in one window, with the two parts it joins and its status.
- Click one to select it in Inventor, then open Inventor's own **Edit Constraint** command from the window.
- Turns **Defer Update** on automatically when more than one constraint is broken, so Inventor does not re-solve after every fix. It turns it off again when you click **Finish**.
- Updates the list when you pick a different constraint (you can switch this off).
- The window stays open while you work, so you can rotate the model and use the browser as normal.

Install without Visual Studio

You do not need Visual Studio or any developer tools. Inventor 2025 or newer already includes the .NET runtime the add-in needs.

1. Get the files

Download `Assembly_Constraint_Urgent_Care.bundle.zip` from this repository's Releases page. It contains one folder, `Assembly_Constraint_Urgent_Care.bundle`, laid out like this:

```
Assembly_Constraint_Urgent_Care.bundle
├── PackageContents.xml
└── Contents
    ├── Assembly_Constraint_Urgent_Care.dll                      <- the add-in itself
    ├── Autodesk.Assembly_Constraint_Urgent_Care.Inventor.addin   <- loads the DLL
    ├── Assembly_Constraint_Urgent_Care.X.manifest
    ├── Assembly_Constraint_Urgent_Care.deps.json
    ├── Assembly_Constraint_Urgent_Care.runtimeconfig.json
    ├── stdole.dll
    ├── Microsoft.VisualStudio.Interop.dll
    └── Microsoft.VisualStudio.Imaging.Interop.14.0.DesignTime.dll
```

Keep all of these files together. The `.dll` and the `.addin` file are the two that matter most. The others are support files the DLL needs.

2. Unblock the download (important)

Windows blocks files that came from the internet, and a blocked DLL will not load in Inventor.

1. Right-click `Assembly_Constraint_Urgent_Care.bundle.zip` and choose **Properties**.
2. At the bottom of the General tab, tick **Unblock**, then click **OK**.
3. Only then extract the zip.

If you already extracted it, unblock the zip and extract again.

3. Copy the folder to Inventor's plug-in location

Close Inventor first. Copy the whole `Assembly_Constraint_Urgent_Care.bundle` folder into **one** of these:

Who can use it	Folder	Notes
Only you (recommended)	`%AppData%\Autodesk\ApplicationPlugins\`	Type this into the Windows Explorer address bar. Create the `ApplicationPlugins` folder if it does not exist.
Everyone on the PC	`C:\ProgramData\Autodesk\ApplicationPlugins\`	Needs administrator rights.

This one location is read by Inventor 2025, 2026 and 2027, so you do not need a separate copy for each version.

When you are done, you should have this file:

```
%AppData%\Autodesk\ApplicationPlugins\Assembly_Constraint_Urgent_Care.bundle\Contents\Assembly_Constraint_Urgent_Care.dll
```

Do not copy only the `Contents` folder, and do not rename the `.bundle` folder's contents. Inventor finds the add-in through `PackageContents.xml`, so it must sit directly inside the `.bundle` folder.

4. Start Inventor

Open an assembly, click the **Assemble** tab, and look for the **Constraint Urgent Care** button at the end of the **Relationships** panel. If Inventor asks whether to trust or load the add-in, allow it.

Using the add-in

Open an assembly and click **Constraint Urgent Care**. The window stays open while you work.

1. The window lists every broken constraint. If there is more than one, **Defer updates** switches on.
2. Click a constraint in the list. It is selected in Inventor.
3. Click **Edit** to open Inventor's Edit Constraint dialog for it. Fix it and click OK in that dialog.
4. Click the next constraint, and repeat. Click **Refresh** at any time to update the list.
5. Click **Finish** when you are done. This turns Defer Update off and updates the assembly once.

Control	What it does
The list	One line per broken constraint: name, the two parts it joins, and its status. The window grows to show 5 to 10 lines, then scrolls.
Edit	Selects the constraint and runs Inventor's **Edit Constraint** command. If that command is not available in your version, select the constraint in the list, then right-click it in the Inventor browser and choose **Edit**.
Refresh	Updates the assembly and rebuilds the list. Constraints you have fixed drop off it.
Finish	Turns Defer Update **off**, updates the assembly once, and closes the window. Closing the window with the X does the same.
Defer updates	Shows whether Defer Update is on (black) or off (gray). It is switched on for you when more than one constraint is broken, and you can tick or untick it yourself. Hover over it for a short explanation.
Refresh on select	When ticked, clicking a different constraint first updates the assembly and rebuilds the list. Untick it if that update brings back Inventor's constraint warning.

Tips:

- Defer Update is normally **off** in Inventor (**Tools > Application Options > Assembly > Defer Update**). The add-in turns it on only while it is needed. **Finish always turns it off**, so if you normally keep it on, tick the option again yourself afterwards.
- The window belongs to the assembly that was active when you opened it. If you open it from a different assembly, the old window closes and a new one opens.
- Suppressed constraints are not listed.

Uninstall

1. Close Inventor.
2. Delete the `Assembly_Constraint_Urgent_Care.bundle` folder from the plug-in folder you used above.

To switch it off without deleting it, use **Tools > Add-Ins > Add-In Manager** in Inventor, select **Assembly Constraint Urgent Care**, and untick **Loaded/Unloaded** and **Load Automatically**.

Troubleshooting

There is no Constraint Urgent Care button. - Look at the end of the **Relationships** panel on the **Assemble** tab, with an assembly open. The button is not added for parts or drawings. - Check the folder layout in step 3. `PackageContents.xml` must be directly inside `Assembly_Constraint_Urgent_Care.bundle`. - Make sure you unblocked the zip before extracting (step 2). Right-click the `.dll`, choose **Properties**, and tick **Unblock** if you see it. - Restart Inventor after copying the files. - Open **Tools > Add-Ins > Add-In Manager** and look for **Assembly Constraint Urgent Care**. If it is listed but not loaded, tick **Loaded/Unloaded**. - If a message says the Relationships panel was not found, Autodesk may have renamed or moved it in your version. See Common changes.

A message says to open an assembly. The button only works when the active document is an assembly (`.iam`).

The Edit button does nothing. The window says what it tried. Select the constraint in the list, then right-click it in the Inventor browser and choose **Edit**.

Inventor's constraint warning still appears. Untick **Refresh on select** and use the **Refresh** button when you want to update the list. The add-in cannot always stop Inventor from showing the warning.

The list is empty but a constraint looks wrong. Click **Refresh**. The list shows constraints that Inventor reports as broken or out of date, so a constraint that is only unwanted will not appear.

Two copies of the add-in. If you also built from source, the build installs to the same plug-in folder. Keep only one copy of the bundle.

Build and modify the source code

This section is for people who want to change the add-in for their own use.

What you need

- Windows with **Autodesk Inventor 2025** installed. The project compiles against Inventor 2025's interop library, which also makes the same DLL work on newer versions (see Compatibility).
- **Visual Studio 2022 (17.14 or newer) or Visual Studio 2026** with the **.NET desktop development** workload. The solution uses the newer `.slnx` format.
- The **.NET 8 SDK** (installed with the workload above).

If you do not have Inventor 2025, change `InteropVersion` in `Assembly_Constraint_Urgent_Care.vbproj` to a version you do have.

Build and run

1. Close Inventor.
2. Open `Assembly_Constraint_Urgent_Care.slnx` in Visual Studio.
3. Choose **Build > Build Solution**.
4. The build copies the add-in bundle to `%AppData%\Autodesk\ApplicationPlugins\Assembly_Constraint_Urgent_Care.bundle`. Start Inventor and the add-in loads.
5. To debug, press **F5**. Visual Studio starts the Inventor version named in `DebugInventorVersion` in the project file (2025 by default). Change that number to debug in a different version.
6. **Build > Clean Solution** removes the installed bundle again.

To make a copy for people without Visual Studio, run `Make-Release.ps1` (right-click it and choose **Run with PowerShell**). It builds the Release version without installing it, and writes `release\Assembly_Constraint_Urgent_Care.bundle.zip`, laid out as shown in Get the files. Attach that zip to a GitHub release.

Project map

File	What it is for
`StandardAddInServer.vb`	The add-in's entry point. Adds the **Constraint Urgent Care** button to the **Relationships** panel of the **Assemble** tab in the Assembly environment, and re-adds it if the ribbon is reset.
`Globals.vb`	The Inventor application reference and the add-in's GUID.
`Command Tools\UrgentCare\UrgentCareCommand.vb`	Creates the button and its icons. Runs when the button is clicked: opens the window, or brings it forward if it is already open.
`Command Tools\UrgentCare\UrgentCareForm.vb`	The window: the list, Defer Update handling, Edit, Refresh and Finish. Built in code, with no designer file.
`Utility Tools\IconLoader.vb`, `PictureConverter.vb`	Helpers: load one size from the icon file, and convert it for Inventor's API.
`Resources\ConstraintUrgentCare.ico`	The button and window icon, with 16 and 32 pixel images in one file.
`Autodesk.Assembly_Constraint_Urgent_Care.Inventor.addin`	The manifest Inventor reads to find and load the add-in.
`Assembly_Constraint_Urgent_Care.X.manifest`	Lets .NET register the add-in with COM without writing to the registry.
`PackageContents.xml`	Tells Inventor's plug-in loader where the `.addin` file is.
`Assembly_Constraint_Urgent_Care.vbproj`	The project file: target framework, Inventor versions, install location and the post-build copy.
`Make-Release.ps1`	Builds Release and zips the bundle for a GitHub release.

Common changes

- **Rename the button or change its tooltip:** edit `buttonLabel` and the tooltip text in `CreateButton` in `UrgentCareCommand.vb`.

- **Put the button on a different panel:** edit the `FindPanel(...)` call in `AddToUserInterface` in `StandardAddInServer.vb`. The tab and panel names are the ones shown in Inventor's ribbon.
- **Change the icon:** replace `Resources\ConstraintUrgentCare.ico`. Include at least 16 and 32 pixel images with transparency.
- **Change which Edit command runs:** see the `names` list in `OnEdit` in `UrgentCareForm.vb`.
- **Change how many rows show:** see `ResizeToRows` in `UrgentCareForm.vb` (5 to 10 by default).
- **Change which constraints are listed:** see `Reload` in `UrgentCareForm.vb`. It lists every constraint whose health status is not up to date and not suppressed.

If you copy this into your own add-in

Give your add-in its own **GUID** so it does not clash with this one. Replace the GUID `365B2F23-E672-443B-96AC-252279713C68` everywhere it appears:

1. `Globals.vb` (the `_simpleAddInClientId` constant)
2. `Autodesk.Assembly_Constraint_Urgent_Care.Inventor.addin` (`ClassId` and `ClientId`)
3. `Assembly_Constraint_Urgent_Care.X.manifest` (`clsid`)
4. `PackageContents.xml` (`ProductCode`)

Also rename the assembly and files if you want both add-ins installed at once.

Keep it compatible with newer Inventor versions

- The Inventor interop library is **embedded** in the add-in. That is why a build against Inventor 2025 also runs on 2026 and 2027.
- Autodesk sometimes renames or removes members. This add-in uses `AssemblyOptions.DeferUpdate`, `AssemblyConstraint.HealthStatus`, and the built-in command named `AssemblyEditConstraintCtxCmd`. The command name was confirmed on Inventor 2025.4 and 2027.
- Before using a new Inventor API member, look it up in the `Autodesk.Inventor.Interop.xml` file next to the interop DLL (in `C:\Program Files\Autodesk\Inventor <year>\Bin\Public Assemblies`) for the oldest **and** the newest version you want to support.
- The `.addin` file only loads in Inventor 2025 or newer, because older versions run on .NET Framework and cannot load a .NET 8 add-in.

Visual Basic things to watch for

- Files that use `Imports Inventor` can clash with .NET names such as `Point`, `File`, `Path`, `Environment` and `Color`. Write `System.Drawing.Point`, `System.IO.File`, `System.IO.Path`, `System.Environment` and `System.Drawing.Color` in full.
- XML comments in the `.vbproj` cannot contain two hyphens in a row.

Disclaimer

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It was written for Inventor 2025, 2026 and 2027 and may stop working in other versions.

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